Row-Action Image Reconstruction Algorithm Using ℓ_p-Norm Distance to a Reference Image

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***Abstract*—This work investigates the problem of image reconstruction from small number of projection views in x-ray computed tomography (CT) imaging. The number of acquired projection views has a large influence on accuracy and stability of the image reconstruction problem. However, measuring the projection data over small number of views leads to a patient dose reduction and/or minimization of imaging time which become a principal target in many clinical applications. The presented work aims to develop a row-action type reconstruction algorithm that include *a priori* known information extracted from a reference image. The proposed method is based on the fact that, in many CT imaging applications, some physical and anatomical structures and the corresponding attenuation information of the scanned object can be *a priori* known. The main idea is to include a distance function consisting of ℓ_p norm of the reconstructed image into the cost function for image reconstruction. The constrained minimization problem is then transferred to the corresponding non-constrained maximization dual problem using Lagrangian duality. Experimental results indicate that the proposed reconstruction algorithms can effectively reduce the streak artifacts when a simple reference image is used.**

I. Introduction

In this paper, we investigate the problem of image reconstruction from small number of projection views in x-ray computed tomography (CT) imaging. The number of acquired projection views has a large influence on accuracy and stability of the image reconstruction problem. However, in many imaging applications, the available projection data are limited to only a small number of views less than theoretically required by Shannon-Nyquist sampling theorem [1]. Moreover, measuring the projection data over small number of views, as shown in Fig. 1, leads to a patient dose reduction and/or minimization of imaging time which become a principal target in many clinical applications. It is known that image reconstructed from such limited data suffers from streak/aliasing artifacts that degrade image quality. In literature, several approaches have been proposed to overcome this problem for different clinical applications (e.g. [2]–[5]).

The presented work aims to develop a row-action type reconstruction algorithm that include *a priori* known information extracted from a reference image. The main idea is to include a distance function consisting of ℓ_p norm of the reconstructed image into the cost function for image reconstruction. The constrained minimization problem is then transferred to the corresponding non-constrained maximization dual problem using Lagrangian duality. As detailed later, using the concept of Lagrangian duality enable us to optimize the cost function using a simple gradient approach.

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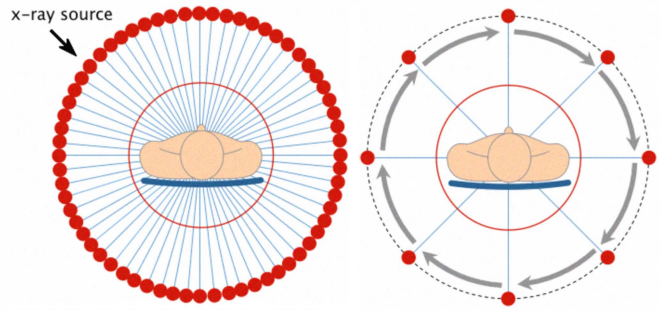

Fig. 1. CT imaging configurations. The positions of x-ray source for data acquisition for full scan (left) and limited views (right).

II. Imaging model

A. Compressed sensing and image sparsity

The recently introduced theory of compressed sensing, becomes an important approach to solve the data limitation problems in tomographic imaging and several other signal processing applications [6], [7]. The main idea is based on the minimization of the cost function that include ℓ_1/ℓ_0 norm of a sparse representation of the target image. It is easy to observe the powerful of ℓ_1/ℓ_0 norm and similar functions in finding the sparse solution from several feasible ones. Hence, it is widely used to solve the under-determinant inverse problems. However, in many medical imaging applications, the target object is non-sparse and additional sparsification transformation such as image gradient or wavelet transform is employed.

In the presented approach, we assume that a reference image that represent normal and expected structures of the scanned object is *a priori* known. When the reference image becomes closer to the reconstructed image, the difference takes the form of sparse image. Let $\vec{x} = (x_1, \ldots, x_n)^\top$ denotes the image vector, we can represent the reconstructed image as the sum of two components. $\vec{x} = \vec{u} + \vec{v}$, where $\vec{u} = (u_1, \ldots, u_n)^\top$ represents the *reference image* and $\vec{v} = (v_1, \ldots, v_n)^\top$ is the *sparse image* which includes unknown regions (such as lesions, tumors, abnormalities,...,etc.) that cannot be expected in prior to the reconstruction. Medical imaging dedicated for

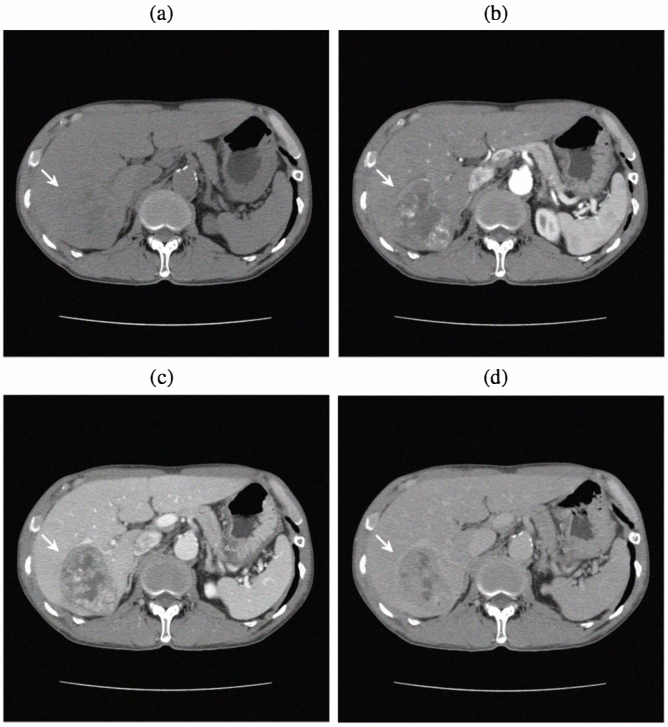

Fig. 2. A single slices of four-phase abdominal CT images of 65 years old patient with liver tumor. Image reconstructed from scan during (a) pre-contrast phase, (b) early phase, (c) portal phase and (d) late phase. Display window is [-130,260]HU.

diagnosis, in many cases, aims to investigate the existence of regions belong to $\vec{v}$ in the imaged object. As the contribution of components $\vec{v}$ to $\vec{x}$ is relatively small compared to that of $\vec{u}$, the image sparisty can be enforced by including the component $\vec{u}$ in the image reconstruction cost function.

B. Create a reference image

The *a priori* knowledge of a reference image can be computed in effective way in several x-ray CT applications. We will discuss a single example that describe how to compute a good reference image in clinical applications. We consider the four-phase abdominal CT imaging. In this imaging protocol, the patient is subject to be scanned four times in pre-contrast, early, portal and later phases. Example of images resulted from this clinical application are shown in Fig. 2. This imaging protocol is important for an effective diagnosis of liver tumors and other diseases in the abdominal region. However, it provides the patient with extremely high dose and uncomfortable long time scan. To reduce the patient dose and imaging time, we proposed the following low-dose protocol. First, we scan the patient with normal dose to obtain the image corresponding to the pre-contrast scan. This image is used to create a reference image to be included in the reconstruction of successive low-dose or small views scans corresponding to the early, portal and later phases. As the four scans are done within the same imaging session, we expect that a simple pre-processing calculations including calibration and image registration can be used to provide a high quality reference image for effective dose reduction in later scans. An illustration of the proposed modification in this imaging protocol is shown in Fig. 3 The fine details of the pre-processing step is not included here and is beyond the scope of this paper.

III. Proposed Method

A. Cost function and minimization strategy

The proposed reconstruction method is based on minimizing the following cost function:

$$\|\vec{x}-\vec{u}\|_p^p/p \text{ subject to } A\vec{x}=\vec{y}, \vec{x}\geq 0, \quad (1<p\leq 2). \quad (1)$$

where $\vec{y}=(y_1,\ldots,y_m)^\top$ is the measured projection data and $A=\{a_{ij}\}$ is the $m\times n$ matrix that model the x-ray CT system. As $m \ll n$, the number of feasible solutions of problem (1) is large. The ℓ_p norm distance between $\vec{x}$ and $\vec{u}$ aims to enforce image sparisty, considering that $\vec{u}$ is relatively close to $\vec{x}$. The optimization problem constraint represents a data fidelity term $A\vec{x}=\vec{y}$ to ensure the consistency between the reconstructed image $\vec{x}$ and the projection data $\vec{y}$. The motivations behind using ℓ_p norm with $(1<p\leq 2)$ instead of ℓ_1/ℓ_0 to solve this problem are: (1) unlike ℓ_1/ℓ_0 norm, the resulting problem is convex, (2) it is possible to maximize the corresponding dual problem with simple gradient methods and (3) when p is close to 1.0, it can achieve a similar performance to ℓ_1 norm in finding the sparse solution. In the results presented here, we have used $p=1.1$.

The cost function in Eq. (1) is optimized using the Lagrangian duality [8], [9]. Consider the following optimization problem with the linear equality constraint:

$$\text{minimize } F_{\vec{u}}(\vec{x}) \text{ subject to } A\vec{x}=\vec{y}, \quad (2)$$

$$F_{\vec{u}}(\vec{x})=\|\vec{x}-\vec{u}\|_p^p/p.$$

The Lagrange dual problem associated with the problem in Eq. (2) is donated as:

$$\text{maximize } D(\vec{\lambda}), \quad (3)$$

$$D(\vec{\lambda})=\inf_{\vec{x}\in\mathbf{dom}(F)}\mathcal{L}(\vec{x},\vec{u},\vec{\lambda}),$$

where $\vec{\lambda}=(\lambda_1,\ldots,\lambda_m)^\top$ is the vector of dual variables (or Lagrange multipliers) with optimal value given by $\vec{\lambda}^*$ and $\mathcal{L}(\vec{x},\vec{u},\vec{\lambda})$ is the Lagrangian function associated with Eq. (2) and is defined as:

$$\mathcal{L}(\vec{x},\vec{u},\vec{\lambda})=F_{\vec{u}}(\vec{x})-\lambda^\top(A\vec{x}-\vec{y}). \quad (4)$$

The conjugate function of $F_{\vec{u}}(\vec{x})$ is then defined as:

$$G_{\vec{u}}(\vec{y})=\sup_{\vec{x}\in\mathbf{dom}(F)}\left(\vec{y}\cdot\vec{x}-F_{\vec{u}}(\vec{x})\right), \quad (5)$$

with prime dual relation defined as $\vec{x}^*=\nabla G(A^\top\vec{\lambda}^*)$, where $\vec{x}^*$ is the minimum value for problem in Eq. (2).

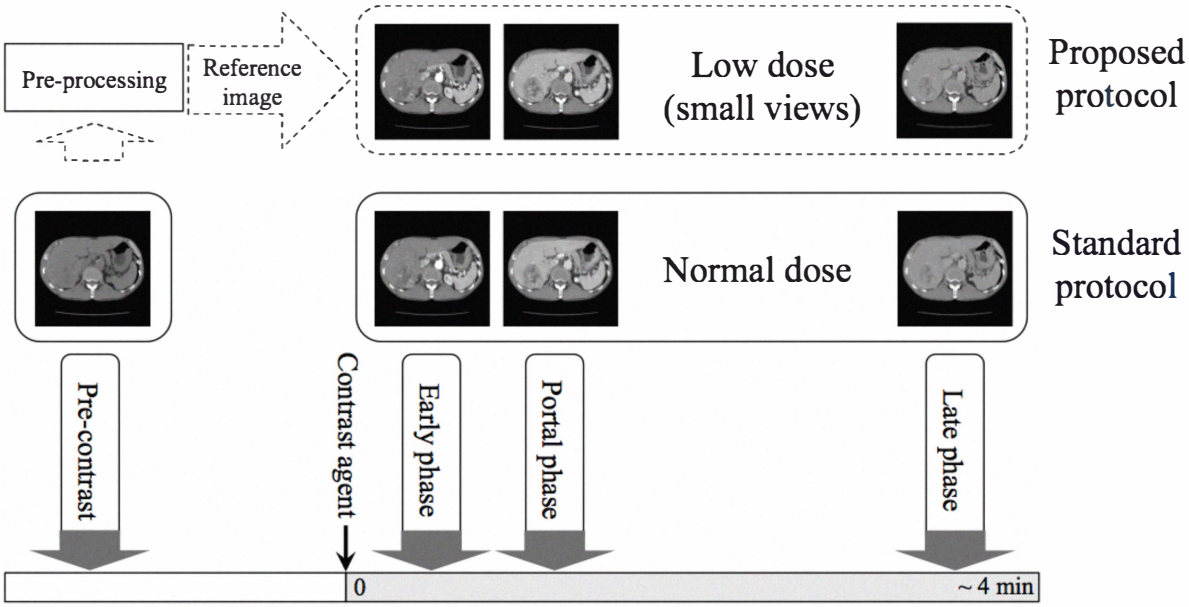

Fig. 3. Four-phase abdominal CT imaging protocol and the proposed modifications to reduce the patient dose.

B. Iterative reconstruction algorithm

The linear program in Eq. (1) can be rewritten as follows:

$$\text{minimize} \sum_{j=1}^{n} \frac{1}{p}|x_j - u_j|^p \text{ subject to } A\vec{x} = \vec{y}, \vec{x} \geq 0, \quad (6)$$

with corresponding dual problem defined as:

$$\text{maximize } D(\vec{\lambda}) = \vec{y}^\top \vec{\lambda} - G_{\vec{u}}(A^\top \vec{\lambda}), \quad (7)$$

where the conjugate function $G_{\vec{u}}$ is defined as:

$$G_{\vec{u}}(\vec{b}) = \sum_{j=1}^{n} g_{u_j}(b_j), \quad (8)$$

$$g_a(t) = \begin{cases} -\frac{1}{p}a^p & (t < -a^{p-1}) \\ \frac{1}{q}|t|^q + at & (t \geq -a^{p-1}) \end{cases}, \quad q = \frac{p}{p-1}.$$

The dual problem in Eq. (7) is solved using the coordinate ascent method to get the optimal value $\vec{\lambda}^*$, then $\vec{x}^*$ is computed using the following primal-dual relation:

$$x_j^* = g'_{u_j}(A^\top \vec{\lambda}^*|_j), \quad (9)$$

$$g'_a(t) = \begin{cases} 0 & (t < -a^{p-1}) \\ \text{sgn}(t)|t|^{q-1} + a & (t \geq -a^{p-1}) \end{cases}. \quad (10)$$

The iterative reconstruction algorithm is implemented as follows. First, we iteratively solve the maximization problem in Eq. (7) to obtain the dual solution $\vec{\lambda}^*$, Then, the primal-dual relation in Eq. (9) is used to find the corresponding solution of the original problem in Eq. (6). The proposed Raw-Action algorithm using Reference Image, which we called RARI, is summarized as follows:

Compute reference image $\vec{u} = (u_1, \ldots, u_n)$
Initialization $w^{(0)} = (0, \ldots, 0)$, $k \leftarrow 0$
repeat
  $\vec{w}^{(k,1)} = \vec{w}^{(k)}$
  for $i = 1 : m$ **do**
    $\vec{w}^{(k,i+1)} = \vec{w}^{(k,i)} + \beta\Delta_i(w_1^{(k,i)}, \ldots, w_n^{(k,i)})\vec{a_i}$
  end
  $\vec{w}^{(k+1)} = \vec{w}^{(k,m+1)}$
  $k \leftarrow k + 1$
until *stopping criterion*;
for $j = 1 : n$ **do**
  $x_j^* = g'_{u_j}(w_j^{(k+1)})$
end

where β is a relaxation parameter and the component Δ_i is the update for the coordinate ascent method used to solve problem in Eq. (7) and is defined as:

$$\Delta_i(w_1, \ldots, w_n) = \frac{y_i - \sum_{j=1}^{n} a_{ij} g'_{u_j}(w_j)}{\max(\Gamma, \sum_{l=1}^{n} a_{il}^2 h_{u_l}(w_l))}, \quad (11)$$

$$h_a(t) = \begin{cases} 0 & (t < -a^{p-1}) \\ \frac{1}{2}(q-2)a^{2-p} & (t = -a^{p-1}) \\ (q-2)|t|^{q-2} & (t > -a^{p-1}) \end{cases}.$$

where $h(.)$ is an approximation to the second derivative of $g_a(t)$ as it is undefined at $t = -a^{p-1}$ and Γ is an upper bound constant to avoid the vanishing of the denominator [9].

IV. Experimental Results

A. Simulation study

A simulation study was performed using a modified version of the Shepp-Logan phantom. The target of image recon-

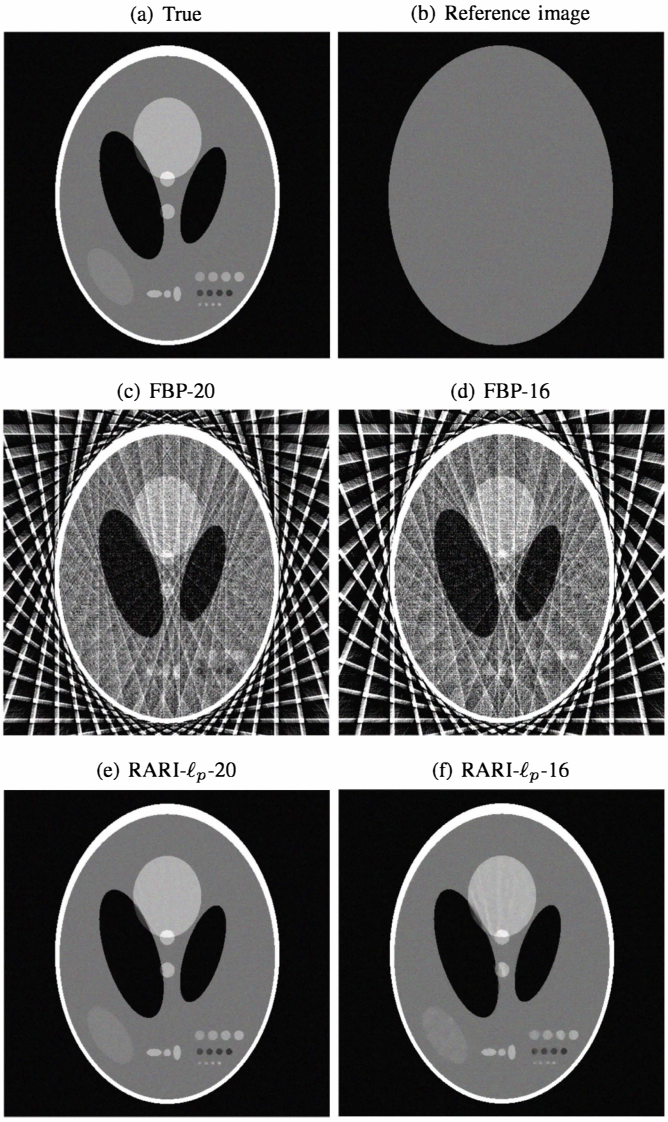

Fig. 4. Reconstruction results of the modified Shepp-Logan phantom in (a). Image reconstructed from 20 and 16 projection views using standard FBP in (c) and (d) and using RARI in (e) and (f), respectively. The reference image used for RARI method is in (b).

struction here is to reduce the image streak artifacts without losing the contrast of the insert spots. An object matrix of 512×512 pixels is projected over 512(bins) and 20/16 (views) over $180°$ with parallel-beam geometry. The reference image $\vec{u}$ is assumed to be a uniform image registered perfectly with the true object, as shown in Fig. 4(b). The simulation was performed through 500 iterations with Γ =256 and β =0.4 and 1.0 for $m = 20$ and 16, respectively. Reconstructed images from the proposed method and the conventional Filtered Backprojection (FBP) are shown in Fig. 4. It is clear from these simulated data that the proposed method can significantly suppress streak artifacts when the projection data is measured over small number of projection views (16 projections) by using a simple uniform reference image. However, it is worth noted that the high image sparsity which we can achieve in simulated data using digital phantoms is practically difficult to be available when we move to the case of real data.

B. Pseudo real data

The proposed method was applied to a pseudo real data for abdominal CT scans of two patients. As the original data is not available, we have used a previously reconstructed CT volumes from medical image database [10] and the projection data was created by a forward-projection of a single image slice. The original data were scanned using x-ray power of 120KeV and reconstructed in volumes with slice size of 512×512 pixels with $0.63mm$ pixel size. The image corresponding to the pre-contrast phase is used as a reference image to reconstruct the image corresponding to the portal-phase from small number of projection views. Image reconstruction from 100 projections was implemented with 10 iterations, $\Gamma = 128$, and $\beta = 0.1$. This experiment was repeated for data obtained from two patients. Results are compared with the original reconstruction from full data for a quality assessment. As shown in Fig. 5, image reconstructed from small number of projection views using FBP algorithm suffers from streak artifacts. However, when the information of the reference image is included into the reconstruction using RARI-ℓ_p algorithm, the image quality is enhanced and could be compared to original image reconstructed from full projection data.

V. Conclusion

We proposed a new row-action image reconstruction algorithm from small number of projection views. The main approach is based on minimizing the ℓ_p norm distance to *a priori* known reference image. The constrained minimization problem is converted to unconstrained maximization using Lagrangian duality. The resulting problem is much easier to solve with $p = 1.1$. The proposed algorithm has a large possibilities to be used in clinical application for the purpose of patient dose reduction. We have presented a modification to four-phase abdominal CT scan for significant reduction of ionization radiation. The results demonstrate that the proposed method is superior in preserving image quality even for small and low-contrast structures, when the projection data is highly under-sampled. Future work include a more sophisticated analysis for quantitative assessment of the proposed method. It is also interesting to investigate the possibilities to use the proposed method in solving other data limitation problems such as limited-angle problem.

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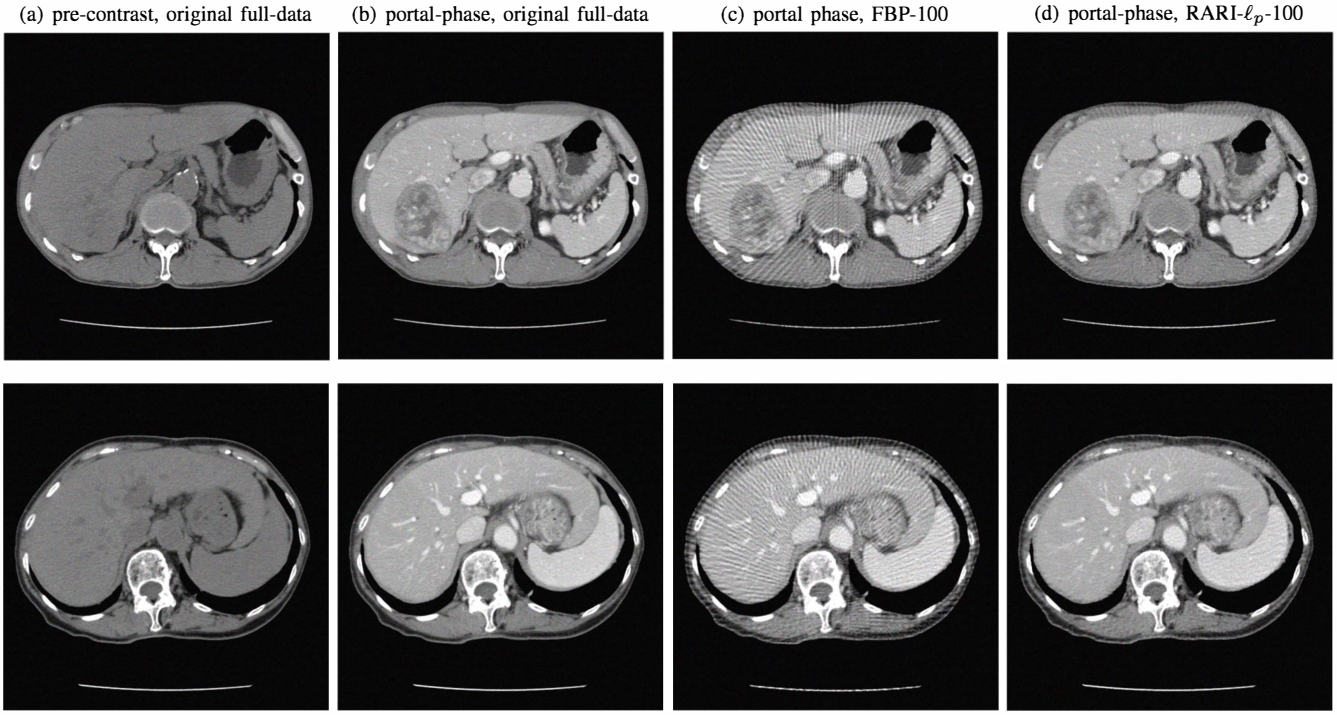

Fig. 5. Reconstruction of abdominal CT scan of two patients in separate rows. (a) original reconstruction corresponding to pre-contrast scan, used as a reference image, and reconstruction corresponding to portal phase using (b) original reconstruction from full data, (c) FBP reconstruction from 100 projection views, and (d) RARI-ℓ_p reconstruction from 100 projection views.

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